

EXPLANATIONS

1. The two components of the peripheral nervous system are:

Somatic nervous system (½ Mark)

Autonomic nervous system (½ Mark)

2. The electric current flowing through the wire is given by the formula:

$$\text{Current} = \frac{\text{Charge}}{\text{Time}}$$

Substituting the given values, we get:

$$\text{Current} = \frac{150 \text{ C}}{60 \text{ s}} = 2.5 \text{ A}$$

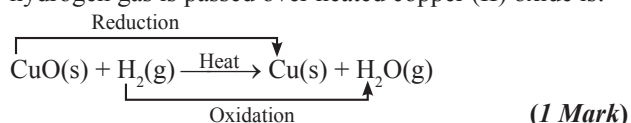
Therefore, the electric current flowing through the wire is 2.5 A. (1 Mark)

3. Out of the Syllabus

4. An aqueous solution of sodium sulphate is neutral because it is a salt of a strong acid (sulphuric acid) and a strong base (sodium hydroxide). When this salt is dissolved in water, it dissociates completely into sodium and sulphate ions, which do not exhibit acidic or basic properties. (1 Mark)

On the other hand, when sodium carbonate is dissolved in water, it is hydrolyzed to some extent and forms sodium hydroxide and carbonic acid. Sodium hydroxide is a strong base. Carbonic acid, on the other hand, is a weak acid. As a result, the mixture contains more hydroxide ions than hydrogen ions, making it basic in nature. (1 Mark)

5. The balanced chemical equation for the reaction when hydrogen gas is passed over heated copper (II) oxide is:



In this reaction, copper (II) oxide is reduced to copper, while hydrogen gas is oxidized to water. (1 Mark)

6. Herbivores have longer small intestines than carnivores due to their plant-based diet. The longer small intestine provides more time for the breakdown of cellulose and other complex carbohydrates in the plant matter and more surface area for the absorption of nutrients. (2 Marks)

7. (i) Lemon is used for restoring the shine of tarnished copper vessels because it contains citric acid, which is a weak acid. Citric acid reacts with the green-coloured basic copper carbonate that forms on the surface of tarnished copper vessels, dissolving it and reveals the shiny copper underneath. (1 Mark)

- (ii) A metal sulphide is converted into its oxide to extract the metal from the sulphide ore because it is easier to obtain pure metal from its oxide, as compared

to its sulphides. The metal sulphide is first roasted in the presence of excess air to convert it into metal oxide and sulfur dioxide gas. The metal oxide is then reduced to obtain the pure metal. (1 Mark)

- (iii) Copper wires are used in electrical connections because copper is a good conductor of electricity. It has low resistance, which means that electric current can flow through it easily. (1 Mark)

8. (i) $\text{ZnCO}_3\text{(s)} \rightarrow \text{ZnO(s)} + \text{CO}_2\text{(g)}$ - Decomposition Reaction (½ Mark)

- (ii) $\text{Pb(s)} + \text{CuCl}_2\text{(aq)} \rightarrow \text{PbCl}_2\text{(aq)} + \text{Cu(s)}$
- Displacement Reaction (½ Mark)

- (iii) $\text{NaBr(aq)} + \text{AgNO}_3\text{(aq)} \rightarrow \text{AgBr(s)} + \text{NaNO}_3\text{(aq)}$
- Double Displacement Reaction (½ Mark)

- (iv) $\text{H}_2\text{(g)} + \text{Cl}_2\text{(g)} \rightarrow 2\text{HCl(g)}$
- Combination Reaction (½ Mark)

- (v) $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Fe}$
- Displacement Reaction

- (vi) $3\text{H}_2\text{(g)} + \text{N}_2\text{(g)} \rightarrow 2\text{NH}_3\text{(g)}$ - Combination Reaction

- (vii) $\text{CaCO}_3\text{(s)} \xrightarrow{\text{Heat}} \text{CaO(s)} + \text{CO}_2\text{(g)}$
- Decomposition Reaction (1 Mark)

9. (i) Dry HCl gas does not change the colour of dry blue litmus paper because HCl gas does not have any H^+ ions in it. As the blue litmus paper only changes its colour in the presence of H^+ ions, it remains unaffected by dry HCl gas. (1 Mark)

- (ii) Alcohol and glucose contain hydrogen but do not conduct electricity because they do not have free electrons or ions to conduct electricity. The hydrogen atoms in these compounds are covalently bonded to other atoms and do not ionize in water to form H^+ ions and free electrons, which are required for electrical conductivity. (1 Mark)

- (iii) The concentration of H_3O^+ ions in a solution of acid decreases when it is diluted because dilution increases the volume of the solution and thus decreases the number of H^+ ions per unit volume. (1 Mark)

10. (i) Digestion of food in the stomach: During digestion, the hydrochloric acid and digestive enzymes in the stomach break down complex food molecules into simpler molecules that can be absorbed by the body. This process involves the breaking down of large food molecules into smaller molecules, which is a characteristic of decomposition reactions. (1 Mark)

(ii) Combustion of coal in the air: In the case of coal combustion, the coal reacts with oxygen to produce carbon dioxide. This reaction is combination reaction which proceeds via the release of heat which means it will be exothermic in nature. **(1 Mark)**

(iii) Heating of limestone: This is a thermal decomposition reaction. In a thermal decomposition reaction, a compound breaks down into simpler substances when heated. In the case of limestone, the limestone breaks down into calcium oxide (Quick lime) and carbon dioxide when heated. This reaction is a combination reaction. Which proceeds via the emission of heat which means it will be exothermic in nature. **(1 Mark)**

11. The rate of breathing in aquatic organisms is much faster than that seen in terrestrial organisms because water contains less oxygen than air. Aquatic organisms must breathe more rapidly to extract enough oxygen from the water to support their metabolic needs. In contrast, terrestrial organisms can extract more oxygen from the air with each breath due to the higher concentration of oxygen in the air. **(2 Marks)**

The pathway of air from the nostrils to the lungs in human beings:-

Nostrils → pharynx → larynx → trachea → alveoli/lungs. **(1 Mark)**

12. Characteristic features of hormonal secretions in human beings.

1. Hormonal secretions in human beings are organ-specific and work on specific targets. **(1 Mark)**
2. They cause chemical changes at the target site, which can affect a wide range of physiological processes. **(1 Mark)**

3. Hormonal secretions are secreted into the bloodstream by the endocrine glands and are transported to their target tissues. **(1 Mark)**

13. (a) The purpose of the formation of urine is to remove waste products from the body. Urine is produced by the kidneys and contains excess water, electrolytes, and metabolic waste products such as urea and creatinine. The process of urine formation helps to maintain the balance of fluids and electrolytes in the body, as well as regulate the pH of the blood. **(1½ Marks)**

(b) If there is no tubular reabsorption in the nephrons of the kidney, then all the essential substances such as glucose, amino acids, and electrolytes will be excreted along with the urine. This would result in a significant loss of these essential substances from the body, leading to a disturbance in the balance of fluids and electrolytes in the body. **(1½ Marks)**

14. Given,

Power of each lamp = 40 W

Voltage = 220 V

Power = Voltage × Current

Rearranging the formula, we can find the current required to safely run a single lamp:

$$\text{Current} = \frac{\text{Power}}{\text{Voltage}} = \frac{40\text{W}}{220\text{V}} = 0.18 \text{ A} \quad (1 \text{ Mark})$$

To find the number of lamps that can safely run on a line of 5 A, we divide the maximum current by the current required for each lamp:

$$\text{Number of lamps} = \frac{5\text{A}}{0.18\text{A}} = 27.77 \quad (1 \text{ Mark})$$

Since we cannot have a fractional number of lamps, we can safely run 27 lamps of rating 40 W: 220 V on this line. **(1 Mark)**

15. Given,

Resistance of a wire, $R = 10 \Omega$

Radius of wire, $r = 0.01 \text{ cm} = 0.01 \times 10^{-2} \text{ m}$

Resistivity, $\rho = 50 \times 10^{-8} \Omega \text{ meter}$

The resistance of the wire is given by the formula:

$$\text{Resistance} = \frac{\rho \times l}{A} \quad \dots(i) \quad (1 \text{ Mark})$$

The area of the wire is given by the formula:

$$\text{Area} = \pi \times (\text{radius})^2$$

$$\text{Area} = 3.14 \times (0.01 \times 10^{-2} \text{ m})^2 = 3.14 \times 10^{-8} \text{ m}^2 \quad (1 \text{ Mark})$$

Substituting all these values in the equation (i), we get:

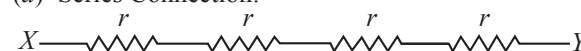
$$10 = \frac{50 \times 10^{-8} \times l}{3.14 \times 10^{-8}}$$

$$l = 0.628 \text{ m}$$

Therefore, the length of the wire is 0.628 m. **(1 Mark)**

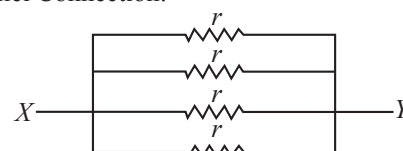
16. Four resistors of r ohm each can be connected in a circuit in the following ways:

(a) Series Connection:



$$\text{Equivalent Resistance} = r + r + r + r = 4r \quad (\frac{1}{2} \text{ Mark})$$

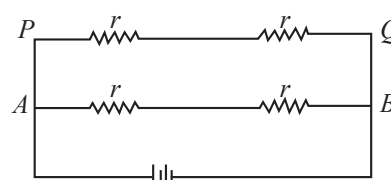
(b) Parallel Connection:



$$\frac{1}{R} = \left(\frac{1}{r}\right) + \left(\frac{1}{r}\right) + \left(\frac{1}{r}\right) + \left(\frac{1}{r}\right) = \frac{4}{r}$$

$$\text{Equivalent Resistance, } R = r/4 \quad (\frac{1}{2} \text{ Mark})$$

(c) Combination of Series and Parallel Connection:



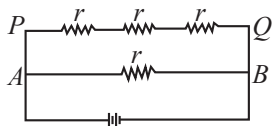
$$\text{Resistance } (PQ) = R_1 = r + r = 2r$$

$$\text{Resistance } (AB) = R_2 = r + r = 2r$$

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{2r} + \frac{1}{2r} = \frac{1}{r}$$

Equivalent Resistance, $R = r$ (½ Mark)

(d) Combination of Parallel and Series Connection:



$$\text{Resistance } (PQ) = R_1 = r + r + r = 3r$$

$$\text{Resistance } (AB) = R_2 = r$$

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{3r} + \frac{1}{r} = \frac{4}{3r}$$

$$\text{Equivalent Resistance, } R = \frac{3r}{4} \quad (\frac{1}{2} \text{ Mark})$$

The equivalent resistance of the combination is

(i) Maximum in the series connection (circuit a).

(ii) Minimum in the parallel connection (circuit b).

(1 Mark)

17. (i) Amit can conserve electricity by utilizing energy-efficient electrical appliances. Replacing traditional filament electric bulbs with CFL bulbs and tube lights can help in achieving this. Furthermore, he can opt for a solar water heater instead of an electrical geyser.

(1 Mark)

- (ii) Through his efforts in saving electricity, Amit plays a part in mitigating environmental degradation. Most electrical appliances rely on electricity generated from fossil fuels, leading to air pollution. Similarly, hydroelectricity production causes ecological imbalances. By reducing his electricity consumption, he indirectly contributes to a decrease in pollution.

(1 Mark)

- (iii) Moreover, he can make use of solar energy devices like solar cookers, solar water heaters, and solar cells.

(1 Mark)

18. Out of the Syllabus

19. (a) Corrosion: It is the gradual destruction of a metal by chemical or electrochemical reaction with its environment. It is a natural process that occurs over time, and it can be accelerated by factors such as moisture, oxygen, and acids.

(1 Mark)

- (b) The corrosion of iron is called rusting. Rust is a reddish-brown flaky substance that forms on the surface of iron when it is exposed to moisture and oxygen.

(1 Mark)

- (c) The corrosion of silver is recognized by the formation of a black coating on its surface. The black coating is formed due to the reaction of silver with hydrogen sulphide gas present in the air.

(1 Mark)

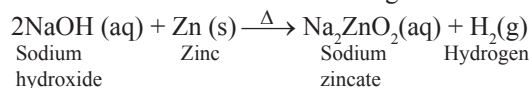
- (d) Corrosion of iron is a serious problem because it can weaken and eventually destroy iron objects. This can lead to structural failures, such as the collapse of bridges or buildings. Corrosion can also be costly, as it requires repairs or replacements of damaged objects.

(1 Mark)

- (e) Methods to prevent corrosion are: It can be prevented by painting, oiling, greasing, galvanising, chrome plating, anodizing or making alloys.

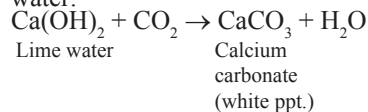
(1 Mark)

20. (i) NaOH solution is heated with zinc granules:



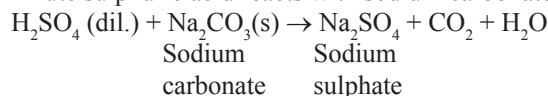
(1 Mark)

- (ii) Excess of carbon dioxide gas is passed through lime water:



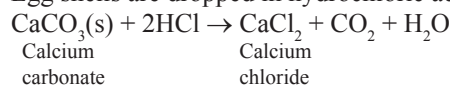
(1 Mark)

- (iii) Dilute sulphuric acid reacts with sodium carbonate:



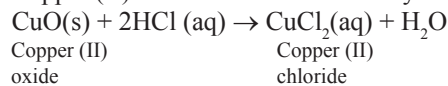
(1 Mark)

- (iv) Egg shells are dropped in hydrochloric acid:



(1 Mark)

- (v) Copper (II) oxide reacts with dilute hydrochloric acid:



(1 Mark)

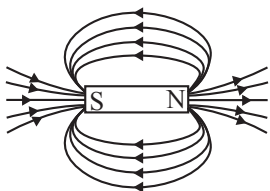
21. (a) The main functions of the nervous system are:

1. Sensory input: Detect changes in the internal and external environment of the body. (1 Mark)
2. Integration: Process and interpret the sensory information to generate appropriate responses. (1 Mark)
3. Motor output: Send signals to muscles and glands to produce a response. (1 Mark)

- (b) In the absence of muscle cells, plants show movement through two types of movements - nastic and tropic movements.

1. Nastic movement: Nastic movement is achieved by changing the amount of water present in the cells of the plant. For example, the diurnal movement of leaves and the response of insectivorous plants to prey. (1 Mark)
2. Tropic movement: Tropic movement is achieved by the action of different hormones in plants. For example, phototropism is achieved by the hormone auxin, while the wilting of leaves is done by the abscisic acid hormone. Growth of the fruits and seeds is achieved by cytokinin. (1 Mark)

22. (a)



(1 Mark)

Two magnetic field lines never intersect each other because if they do, it would mean that at the point of intersection, the direction and magnitude of the magnetic field are different, which is not possible.

(1 Mark)

(b) The power consumed by an electrical appliance is given by the formula:

Power = Voltage $\times$ Current

(1 Mark)

Substituting the given values, we get:

Power = 1.5 kW = 1500 W

Voltage = 220 V

Current = 1500 W / 220 V = 6.82 A

(1 Mark)

Since the current rating of the domestic circuit is 5 A, we expect the circuit breaker to trip or the fuse to blow, resulting in the interruption of the power supply to the electric oven. This is because the current drawn by the electric oven is more than the current rating of the circuit, which is 5 A.

(1 Mark)

23. Resistance of a conductor is the property of a material that opposes the flow of electric current through it.

$$R = \rho \frac{l}{A}$$

(1 Mark)

The SI unit of resistance is ohm (Ω), which is defined as the resistance between two points in a conductor when a constant potential difference of 1 volt produces a current of 1 ampere in the conductor.

(1 Mark)

The resistance of a conductor depends on the following factors:

1. Length of the conductor
2. Cross-sectional area of the conductor
3. Material of the conductor
4. Temperature of the conductor

(1 Mark)

(i) If the length of a wire is doubled, its resistance also doubles. This is because resistance is directly proportional to the length of the conductor.

$$R \propto l$$

(ii) If the radius of a wire is doubled, its resistance reduces to one-fourth of its original value. This is because the cross-sectional area of the wire is proportional to the square of its radius, and resistance is inversely proportional to the cross-sectional area of the conductor.

$$R \propto \frac{1}{A} \propto \frac{1}{\pi r^2}$$

$$\therefore A = \pi r^2$$

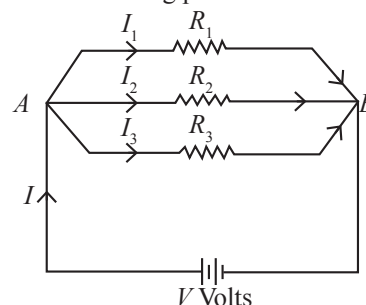
If $r' = 2r$, $A' = 4A$

$$R' = \frac{1}{4} R$$

Hence, If the radius of a wire is doubled, its resistance reduces to one-fourth of its original value.

(2 Marks)

24. (i) Consider the following parallel circuit shown below:



Using Ohm's law, the current through each resistor can be determined as follows:

$$I_1 = \frac{V}{R_1}, I_2 = \frac{V}{R_2}, \text{ and } I_3 = \frac{V}{R_3} \quad (1 \text{ Mark})$$

Let R_p represent their equivalent resistance.

$$\text{Then, we have } V = I \times R_p \Rightarrow I = \frac{V}{R_p}$$

The total current through the circuit is

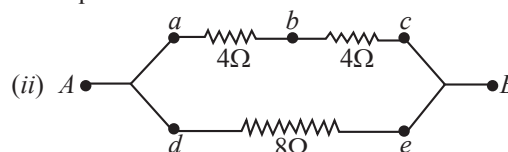
$$I_{\text{Total}} = I_1 + I_2 + I_3$$

$$\therefore \frac{V}{R_p} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3} \Rightarrow \frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\text{Hence, } R_p = \frac{R_1 R_2 R_3}{R_1 R_2 + R_2 R_3 + R_1 R_3}$$

Therefore, the equivalent resistance R_p of a combination of three resistors having resistances R_1 , R_2 and R_3 connected in parallel is given by the above equation.

(2 Marks)



Given, $R_1 = 4\Omega$, $R_2 = 4\Omega$, $R_3 = 8\Omega$

(1 Mark)

R_1 and R_2 are in series combination. So resistance between a and c

$$R_s = R_1 + R_2 = 4 + 4 = 8\Omega$$

Now R_3 is connected in parallel

$$\frac{1}{R_p} = \frac{1}{R_s} + \frac{1}{R_3} = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$$

$$R_p = 4\Omega$$

Therefore, the resistance between points *A* and *B* is 4 Ω .

(1 Mark)

25. (c)

	Zn	Na ₂ CO ₃
HCl	✓	✓
NaOH	✓	—

(1 Mark)

Sodium carbonate and sodium hydroxide both are bases, so do not react with each other.

26. (c) Sodium carbonate and sodium hydroxide both are bases, so do not react with each other. Hence, when they both are mixed, the bases simply get dissociated into their respective ions. (1 Mark)

27. (a) Cu metal is reddish brown (1 Mark)

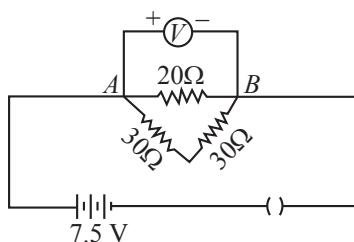
28. (b) Al will displace Fe from its salt solution which results in the formation of colorless solution and grey black deposition of Fe. (1 Mark)

29. (b) This is based on the concept of reactivity series in which more reactive metal will displace the less one from their respective salt solutions. (1 Mark)

30. (c) Correct reading for current *I* and incorrect reading for voltage *V*.

Ammeter has low impedance; voltmeter has high impedance. Voltmeter in parallel with ammeter leads to max current through ammeter, giving accurate reading. Voltmeter not in parallel with resistors, shows incorrect voltage drop. (1 Mark)

31. (b) 0.5 A



Given, $R_1 = 20\Omega$, $R_2 = 30\Omega$, $R_3 = 30\Omega$

R_2 and R_3 are in series combination. So resistance between them

$$R_s = R_2 + R_3 = 30 + 30 = 60\Omega$$

Now R_1 is connected in parallel

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_s} = \frac{1}{20} + \frac{1}{60} = \frac{8}{120}\Omega$$

$$R_{eq} = 15\Omega$$

Therefore, the resistance between points *A* and *B* is 15 Ω .

$$\text{Now, } V = IR_{eq}$$

$$I = \frac{V}{R_{eq}} = \frac{7.5}{15} = 0.5$$

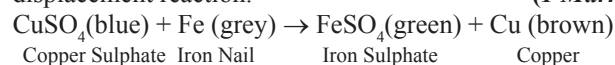
The current recorded by ammeter (*A*) is 0.5 A.

(1 Mark)

32. (a) Pure iodine dissolved in water (1 Mark)

33. (d) In the given experiment, the CO₂ released by the germinating seeds is absorbed by KOH solution that creates a partial vacuum in the flask resulting in the upward movement of water in the tube. Hence, one can conclude that CO₂ is released during the process of respiration. (1 Mark)

34. Iron is more reactive than copper. When an iron nail is placed in a copper sulphate solution, the iron displaces the copper from the solution. The copper deposits on the surface of the iron nail, and the blue colour of the copper sulphate solution fades and the colour of the solution turns green due to the formation of ferrous sulphate. This is a displacement reaction. (1 Mark)

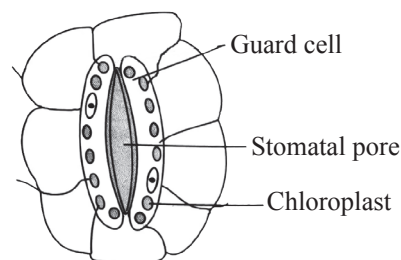


Copper Sulphate Iron Nail Iron Sulphate Copper

(1 Mark)

35. No, the student is not necessarily wrong. The slight difference in the values of resistance could be due to various factors such as the accuracy of the measuring instruments, the presence of stray resistances, the temperature of the resistors, etc. Therefore, the student must ensure that the measuring instruments are calibrated and the resistors are uniform before performing the experiment. (2 Marks)

36.



(2 Marks)